

The smaller scope: building the indicators x sensors x certification crosswalk

Manisha Sarkar for Kevin Xia, 23 September 2026. Item #13 against the existing repository, revised after the call of 22 September

The scope, and the constraint it has to respect

Item #13 on Kevin's list: a three way crosswalk for one species, validated welfare indicators against what commercial PLF sensors actually measure against what certification schemes require and audit, about 25 hours. It answers: which welfare indicators are (a) measurable by AI, (b) acceptable to industry for adoption, and (c) potentially already covered by existing sensors or widely used certification schemes? The valuable cells are indicators already instrumented but required by no scheme, and indicators required but audited only by inspection. Kevin suggested broilers rather than dairy for impact. The decision is open.

The constraint from the call: the mentor's time is one to three hours in total, for strategic direction, not for validating welfare science or reviewing code. The original plan failed that test because it needed rulings I could not make myself. This plan is built so that it needs three decisions from Kevin, on Slack, in about twenty minutes, and nothing else. Every other judgement is made by a written rule and shown both ways in the table, so a reader can disagree with the rule without the table being wrong.

How much already exists

Column	Dairy, in the repository today	Broilers
1. Validated indicators	Done. 57 rows from EFSA 2023 and Welfare Quality, source table cited on every row.	To build from EFSA 2023 Welfare of broilers on farm (EFSA Journal 21(2):7788) and the Welfare Quality poultry protocol. About 4 hours with the same method.
2. What sensors measure	Mostly done. Stygar's 129 products and 18 validated ones with per product validation level; Maroto Molina's technology per measure. Left: map Stygar's validated traits onto the indicator rows, about 4 hours.	The gap. The nearest equivalent is Rowe, Dawkins and Gebhardt-Henrich 2019, A Systematic Review of Precision Livestock Farming in the Poultry Sector, Animals 9(9):614, and a 2025 review by Brassó, Komlósi and Várszegi, Animals 15(4):493. Whether either codes commercial availability and validation status the way Stygar does has to be checked first. If neither does, column 2 would rest partly on vendor claims. About 8 hours including that check.
3. What schemes require	Schema and audit method categories designed. All four standards in the corpus: RSPCA April 2026 with its justification document, FARM v5, GAP v2.0, Certified Humane Edition 23. No rows yet. About 7 hours.	RSPCA broiler standards, the Better Chicken Commitment, GAP chicken, Certified Humane chicken, with the National Chicken Council guidelines as the industry proxy. None in the corpus yet. About 8 hours.
Queries and checks	The three queries are coded. Corpus manifest, link checker, research auditor and pre push hook all work.	Transfer unchanged.

How the three filters are decided without a ruling

- Measurable by AI. An indicator is graded, not flagged: validated commercial (an externally validated product measures it, Stygar), commercial unvalidated (a product claims it, no independent study), research only (Maroto Molina partial), none. The table shows the grade. The 24 EFSA measures with no Welfare Quality equivalent appear as their own tier, labelled, so the reader can include or exclude them.
- Acceptable to industry. Named in the industry authored programme: FARM Animal Care v5 for dairy (NMPF), the National Chicken Council guidelines for broilers. The table says which requirement, quoted, so the proxy is visible and crude on purpose.
- Covered. Required by any of the four schemes, with the audit method coded as inspection, records or sensor accepted, and the requirement text quoted on every row. The two valuable lists fall out of the join by script.

Plan and hours, dairy

Step	Work	Hours
1	Write the three definitions above into the repository and post them on Slack. No reply needed unless Kevin objects.	1
2	Column 2: map Stygar's validated traits and products onto indicator rows with the grade.	4

Step	Work	Hours
3	Column 3: RSPCA with the justification document beside it, 3; FARM v5 with its numeric thresholds, 2; GAP and Certified Humane, 2. One row per indicator per scheme, requirement quoted, threshold exact, audit method coded.	7
4	Run the queries. Produce the table and the two lists.	1
5	Write up, about 1,500 words, plus a one page version in the format of the list entry.	4
6	Two verification passes and corrections.	3
7	Buffer for the extraction judgement calls, the one real risk: when a scheme requirement "names" an indicator.	5
	Total, about three weeks at eight hours a week	25

Broilers: the same seven steps with columns 1 and 2 rebuilt, 30 to 35 hours, and the first week goes to checking whether the poultry reviews support a validated commercial grade. If they do not, the table still works, but column 2 is honest about resting on claims. Everything else transfers: schema, scripts, corpus manifest, verification gate. The list decomposes one species per unit, so dairy now and broilers later is a legitimate sequence, and dairy is the cheaper proof of the method.

How it differs from the existing plan

It is deliverable 2 of the six, kept whole, with the other five dropped: no claims register, no scorecard, no standard text, no logger, no certifier annex. One method change follows: column 2 comes from Stygar's validation results rather than from a claims register, which is better for this question, because measurable by AI should mean measured by a validated product, not claimed. The audience is the one Kevin named on the call, certifiers such as RSPCA; the required but manually audited list is written for them. Nothing built so far is wasted.

What I need from Kevin, on Slack, about twenty minutes

- Species: dairy first, or broilers first. My recommendation is dairy, 25 hours from a standing start, then broilers as the second unit.
- Yes or no: FARM v5 and the NCC guidelines as the industry acceptability proxy.
- Yes or no: the four dairy schemes (RSPCA Assured, FARM v5, GAP, Certified Humane), or the four broiler ones, as the "widely used" set.

Everything else I do myself or route elsewhere. The paywalled paper and a one hour welfare science look at the final shortlist go to the Sentient Futures request help channel, as Kevin suggested. Progress goes on Slack on Fridays as one line and a link, no reply expected.

Deliverables and limits

Deliverables: the crosswalk as a CSV complete for four schemes; the two lists and the full shortlist; a 1,500 word note; a one page entry in the format of the list. Limits, stated in the note: Stygar is 2021 and the market has moved, so the validated set is a floor; the industry proxy is crude and labelled as such; column 3 depends on reading requirement text, which is why every row quotes it and the checker verifies it; and the crosswalk says which indicators could be instrumented, not whether instrumenting them helps the animal.